

LAB ASSIGNMENT 4 REPORT

Cryptanalysis of Shift Cipher Using Brute Force, Dictionary Scoring and Chi-Square Analysis

Submitted By

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Course: Cryptography Laboratory

Programming Language: C++, python

1. Objective

The objective of this assignment is to implement and evaluate cryptanalysis techniques for a Shift Cipher without prior knowledge of the encryption key.

The implemented system performs:

- Brute-force decryption
- Dictionary-based scoring
- Chi-Square statistical analysis

The experimental results are used to compare the key predicted by dictionary scoring with the key predicted by Chi-Square analysis.

2. Introduction

The Shift Cipher is a classical substitution cipher in which every alphabetic character is shifted by a fixed number of positions in the alphabet.

For a key K , encryption is performed using:

$$C=(P+K)\text{mod}26$$

and decryption is performed using:

$$P=(C-K)\text{mod}26$$

where:

- P = plaintext character
- C = ciphertext character
- K = encryption key

The Shift Cipher has only **26 possible keys**, including key 0. This makes exhaustive key searching computationally inexpensive and makes the cipher unsuitable for modern secure communication.

In this assignment, the ciphertext is analyzed using multiple approaches to determine the most probable encryption key.

3. Implementation Details

The complete implementation was developed in **C++** using the **g++** compiler.

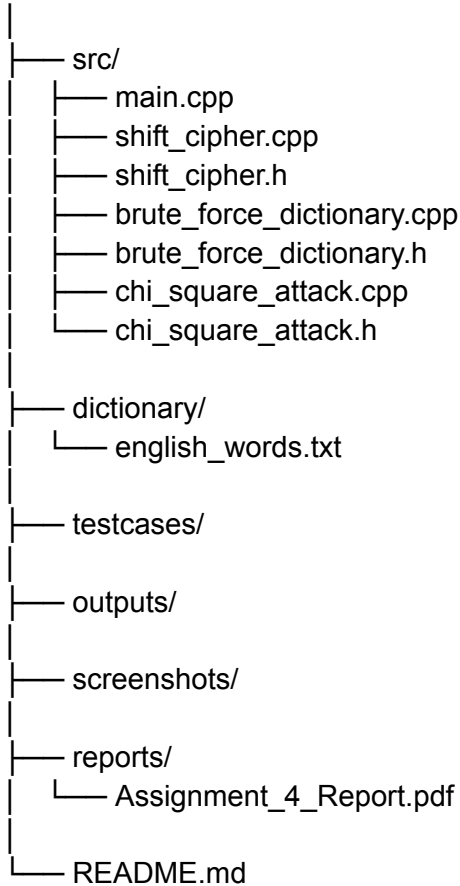
The implementation is divided into separate modules for encryption/decryption, dictionary scoring, and Chi-Square analysis.

3.1 Technologies Used

Component		Details
Programming Language	C++	
Compiler	g++	
Operating Environment	Windows / WSL Ubuntu	
Cipher	Shift Cipher	
Dictionary	English word list	
Cryptanalysis	Brute Force, Dictionary Scoring, Chi-Square	

3.2 Project Structure

shift_cipher_attack/



4. System Implementation

The application consists of three major cryptanalysis components.

4.1 Shift Cipher Module

The Shift Cipher module provides:

- Encryption
- Decryption
- Key handling
- Brute-force generation of all possible decryptions

The key is normalized to the range 0–25.

4.2 Dictionary Scoring Module

The dictionary module loads English words from:

dictionary/english_words.txt

For every possible key, the ciphertext is decrypted and the resulting plaintext is checked against the dictionary.

The number of valid dictionary words is used as the score.

The key producing the highest score is selected as the dictionary-based prediction.

4.3 Chi-Square Module

The Chi-Square module analyzes the frequency of letters in each candidate plaintext.

The observed letter frequencies are compared with standard English letter frequencies using:

$$\chi^2 = \sum [(O_i - E_i)^2 / E_i] \text{ for } i=1 \text{ to } 26$$

where:

- O_i = observed frequency
- E_i = expected English frequency

The key producing the **lowest Chi-Square value** is selected as the predicted key.

5. Test Cases

The implementation was tested using different plaintexts and encryption keys to evaluate the performance of the two automated attacks.

Test Case	Plaintext	Actual Key
TC01	the quick brown fox jumps over the lazy dog	3

TC02	cryptography protects information using mathematical techniques and secret keys	7
TC03	security depends on confidentiality integrity authentication and availability	11
TC04	a secure communication system protects messages from unauthorized access and modification	15
TC05	meet me	8
TC06	this is a test message and this is another test message	19

Each plaintext is first encrypted using the actual key. The resulting ciphertext is then supplied to the cryptanalysis program.

6. Screenshots

They are uploaded on github in the screenshots folder.

7. Conclusion

The Shift Cipher cryptanalysis system was successfully implemented in C++ using brute-force search, dictionary scoring, and Chi-Square analysis.

The experiment demonstrates that the Shift Cipher is vulnerable because of its extremely small key space. Brute-force analysis can generate every possible decryption, while dictionary scoring and Chi-Square analysis provide automated methods for identifying the most probable plaintext.

The experimental comparison of dictionary scoring and Chi-Square analysis shows how different cryptanalysis techniques can produce different predictions depending on the characteristics and length of the ciphertext.

Overall, the assignment provided practical experience in implementing classical cryptanalysis techniques and evaluating their effectiveness against a Shift Cipher.

